

3D PRINT ERROR SOLUTION

NOpaghetti

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3D Printer Industry Status & Outlook

Market Size

\$21.1B in 2023
2032 \$118.9B USD
Projected

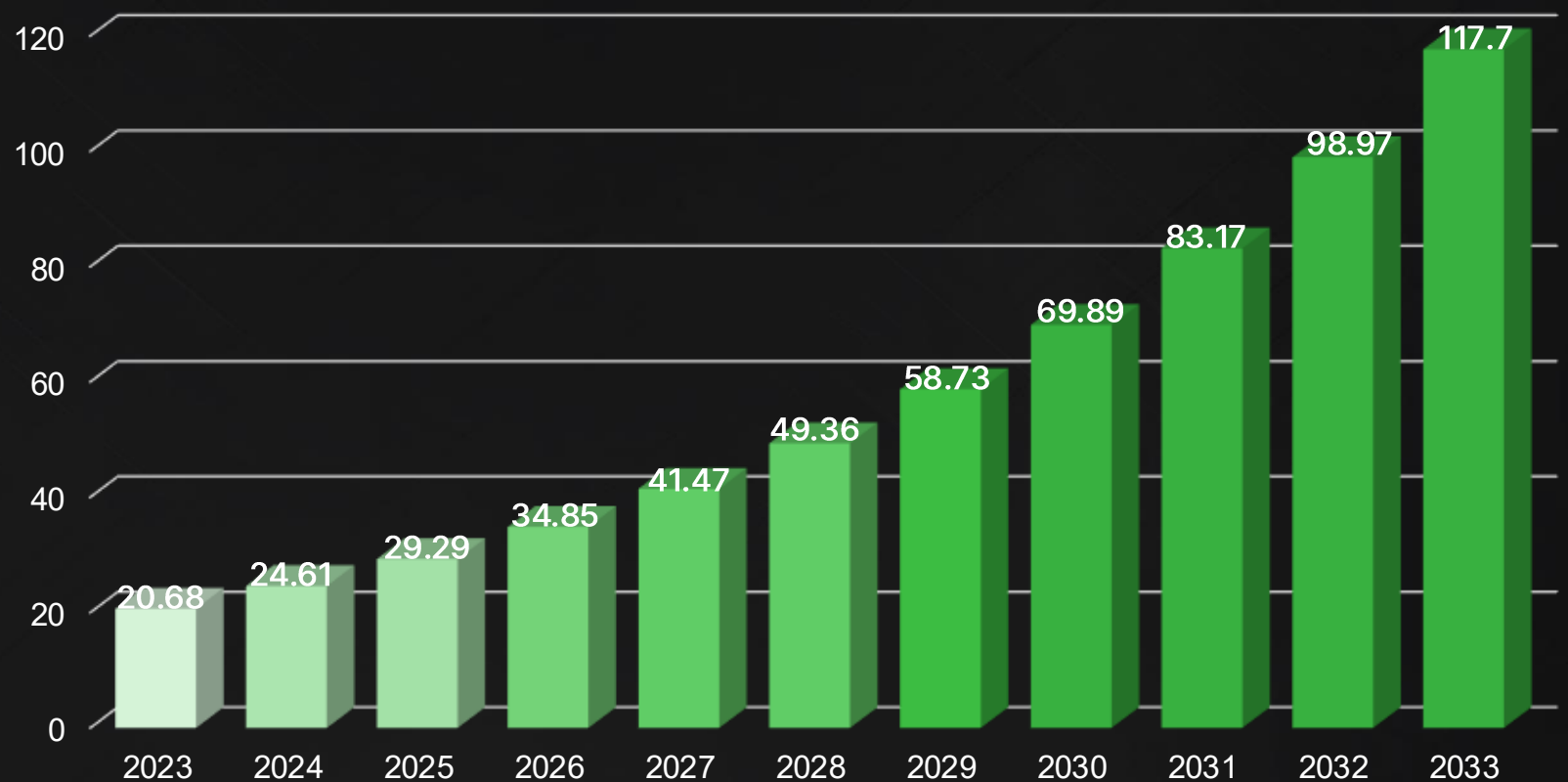
Industry Status

Falling costs fuel parts &
prototyping adoption

Rapid Growth Forecast

2024–2032
~19% CAGR Expected
(Exceptionally High)

3D Printer Market Size Forecast (Unit: USD Billion)



OVERVIEW

Reason for Topic Selection



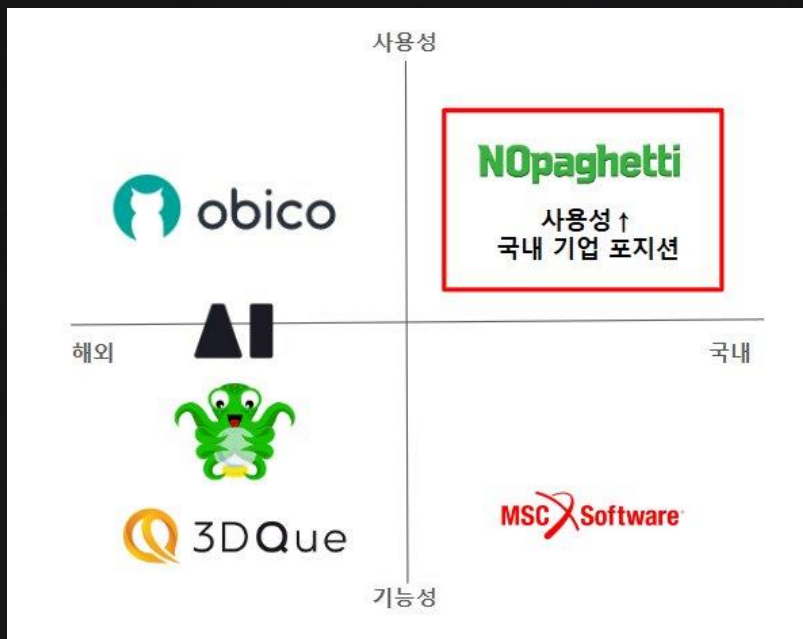
- 2021 HSSMI (UK Mfg. Consultancy) Case Study: Reports 3D Printing **Outputs Discarded at 33%** (~379,000kg waste per 1,148,400kg consumed)
- 80% of 3D printing waste **is caused by print failures**
- 3D print errors waste significant time and electricity Early error detection **improves UX & reduces carbon emissions**

We selected this topic to prevent resource waste by catching 3D print errors, enabling eco-friendly manufacturing and contributing to society through this project.

OVERVIEW

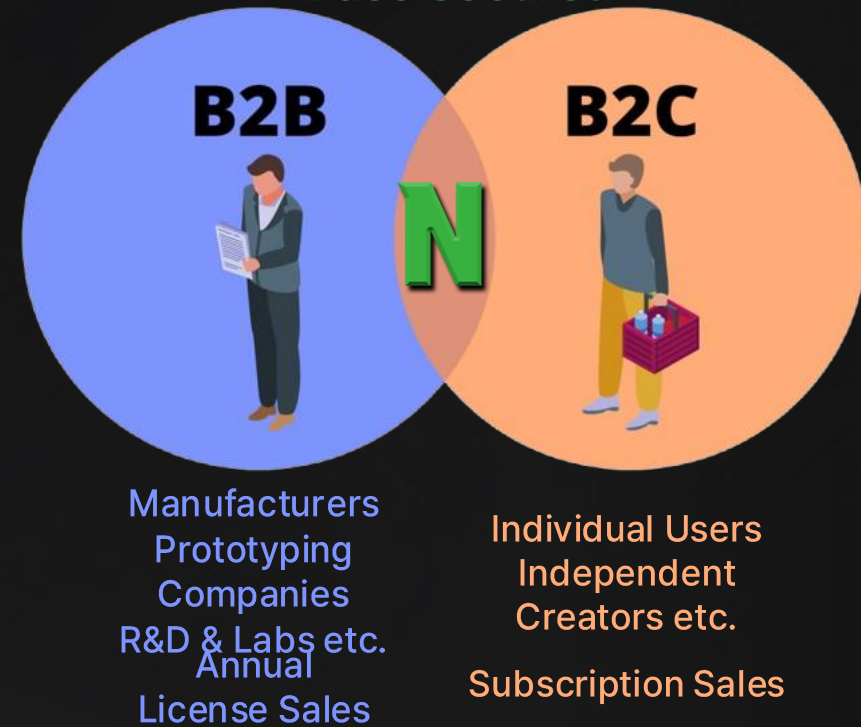
Market Strategy

Domestic Market First-Mover



- Overseas programs → **Korean users face usability challenges**
- Korean-language customer support, community driven info sharing & feedback
- New market entry strengthens **market position**

Via B2B & B2C: Diverse Customer Base Secured



Dual-targeting strategy enhances brand image

Team Members & Roles



Junsang Lee
Project Manager



Duhyeon Nam
Frontend Engineer



Jooyoung Hur
Backend & Server
Engineer



Yujin Lee
Data & ML Engineer



Wonhyeong Son
Market & Data
Analyst



Taehwan Choi
Presentation
Designer

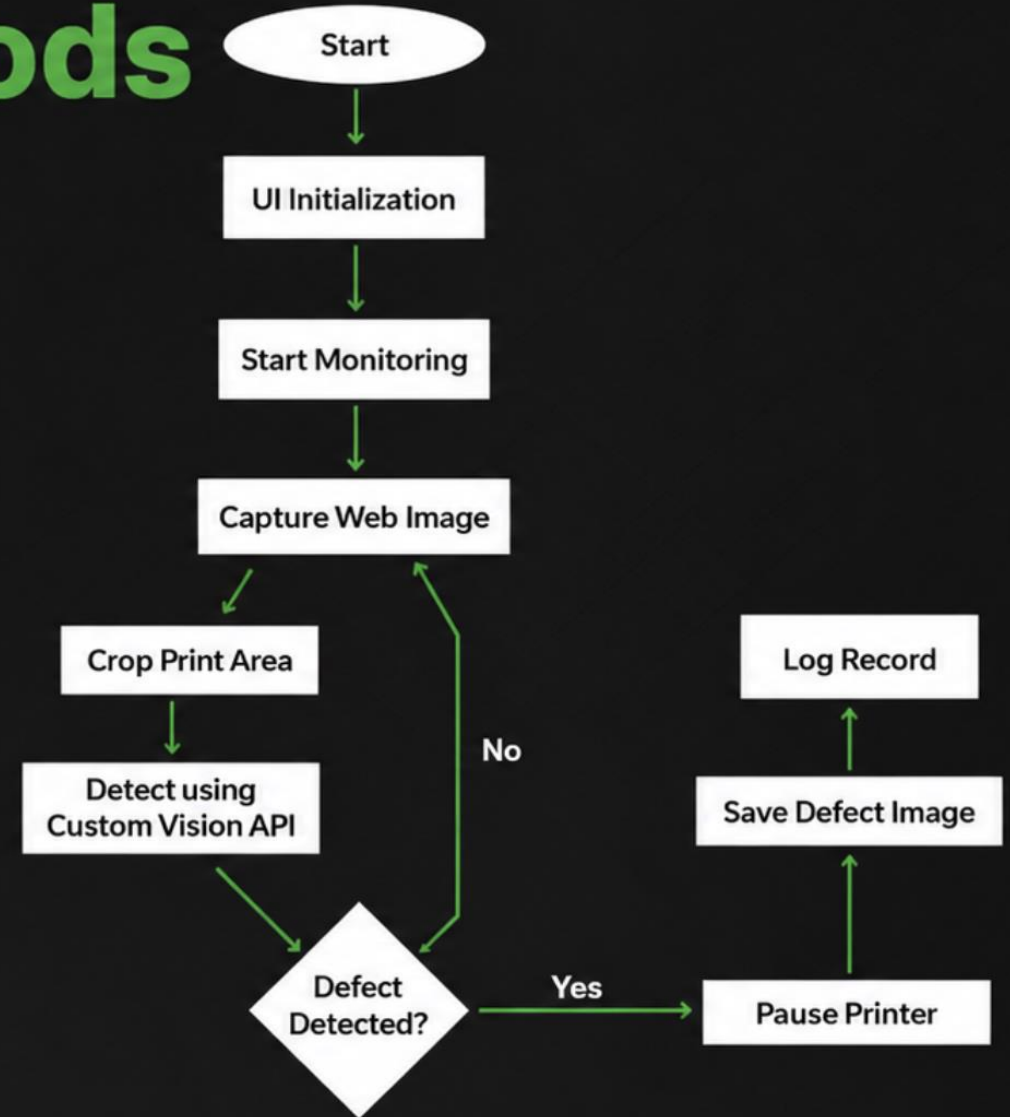
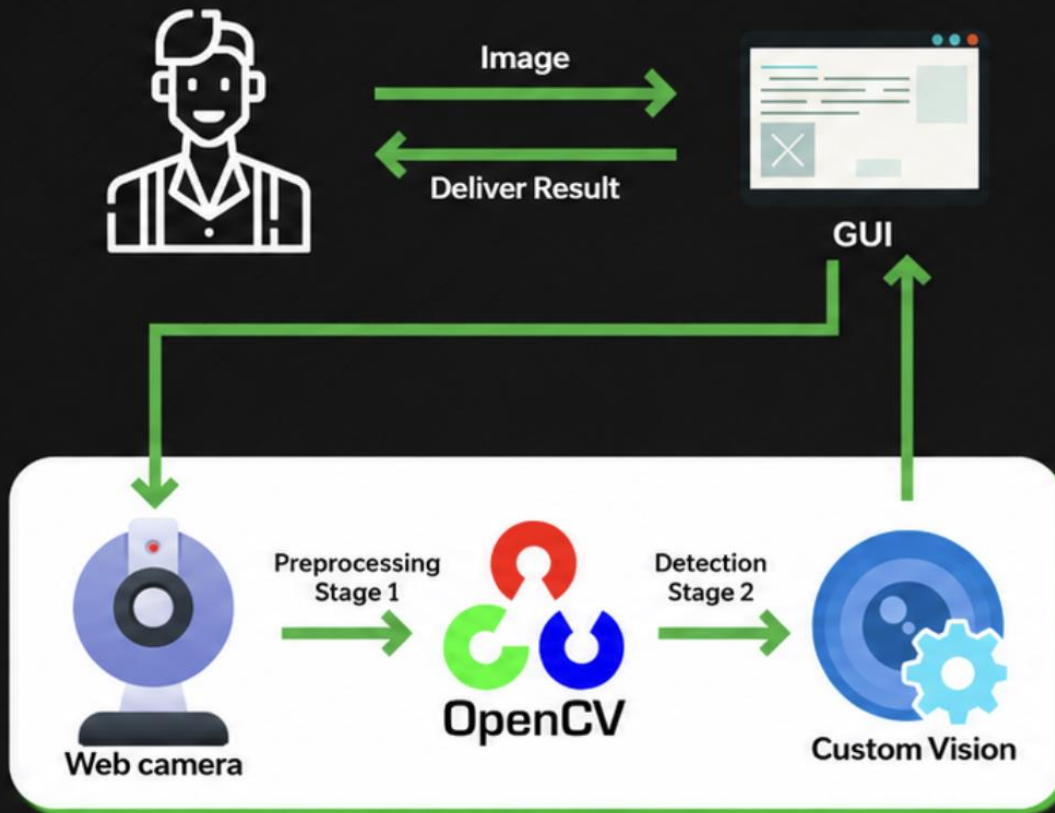
TIMELINE

Project Timeline

10/28 (Mon)	10/29 (Tue)	10/30 (Wed)	10/31 (Thu)	11/1 (Fri)
			Topic Selection	Initial Planning Data Collection
11/4 (Mon)	11/5 (Tue)	11/6 (Wed)	11/7 (Thu)	11/8 (Fri)
Custom Vision Tagging/Classification Research Summary	PPT Design & Writing App UI Implementation	Defect Detection Implementation Control Feature Implementation	Code Team Review PPT Final Review Code Refactoring	Presentation Rehearsal

PROGRESS

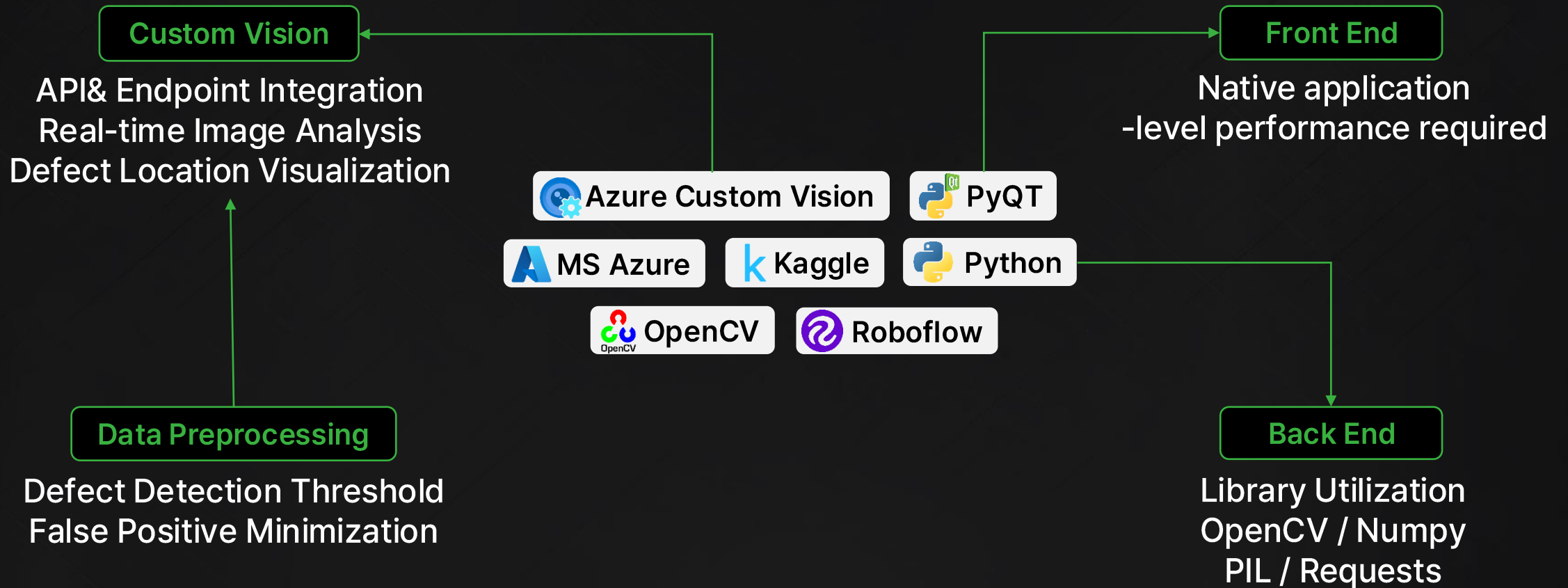
Procedure & Methods



PROGRESS

Procedure & Methods

Tech Stack & Platforms



PROGRESS

Code Team Review

The screenshot displays a VS Code Code Team Review session. The left pane shows a Python file named `version1.py` with the following code:

```
1 import sys
2 import cv2
3 import numpy as np
4 import requests
5 import time
6 from PyQt5.QtWidgets import *
7 from PyQt5.QtCore import *
8 from PyQt5.QtGui import *
9 from azure.cognitiveservices.vision.customvision.prediction import CustomVisionPredictionClient
10 from msrest.authentication import ApiKeyCredentials
11 from PIL import Image
12 import io
13 import os
14 from datetime import datetime
15 from styles import STYLE_SHEET

16
17 class PrintMonitorApp(QMainWindow):
18
19     # 모든 UI 및 변수 초기화를 담당하는 생성자 메소드.
20     def __init__(self):
21         super().__init__()
22         self.setStyleSheet(STYLE_SHEET)
23
24         # URL 설정
25         self.moonraker_base_url = "http://3dro.kr:3002"
26         self.webcam_url = "http://3dro.kr:3002/webcam/"
27
28         # 세션 재사용을 위한 requests 세션 초기화
29         self.session = requests.Session()
30         self.session.headers.update({
31             'User-Agent': 'Mozilla/5.0',
32             'Accept': 'image/jpeg, image/png, */*',
33             'Referer': 'http://3dro.kr:3002/cam'
34         })
35
36         # Azure Custom Vision API 설정
37         self.prediction_key = "71r0X0ueuMkSZLK097Cq98RLNgT6p0KagR6aaymfYTBWkmPFAC1QQJ99AKAYeBjFXJ2w3AAATa
38         self.project_id = "224ff5eb-7998-4705-a4fe-10b7be2d3e4b" #03ad7aee-92b9-4dce-a262-0d3510907126
39         self.iteration_name = "Iterations"
40
41         # 프린터 상태 초기화
42         self.printer_status = "idle"
43
44         # 상태 플래그 초기화
45         self.monitoring = False # 모니터링 실행 상태
46         self.defect_detected = False # 결함 감지 상태
47         self.defect_logged = False # 결함 로그 출력 상태
48         self.last_processed_time = 0 # 마지막 처리 시간
49         self.processing_threshold = 0.5 # 처리 주기 (초)
50
51         # 이미지 처리 관련 변수 초기화
52         self.last_frame = None # 마지막 프레임 저장
53         self.default_image = None # 기본 이미지 지정
```

The right pane shows a video call with participants and a chat window. The chat window contains the following messages:

- 오후 1:13 녹음/녹화가 중지되었습니다. 녹음/녹화 저장 중...
- 오후 1:13 최태현에서 클라우드에 기록 시작
- 기록 최대량 4h
- 이 녹음/녹화에는 암호가 없습니다. 로그인(로그인)을 클릭하여 보거나 변경합니다. 자세한 정보
- 김현철 오후 1:32 프로젝트 공지사항 안내드립니다
- 중요한 내용이나 꼭 읽어보시길 바라며, 여러분께서 작성해주셔야 하는 내용도 있습니다.
- 공지사항 채널 확인하시거나, 아래 링크로 확인해주세요
- 링크: 프로젝트 공지사항
- 최태현 오후 1:35 공지 요약:
- 1) 11월 8일 금요일 낮 12시 기준 산출물 올단에 있는 시연 영상, 태일리 리포트 등 항목들로 평가
- 2) 우리팀은 총 15팀 중 5번째 순서
- 3) 발표시간 15분으로 확장, 10분에 맞춰서도 감점 없음
- 메시지를 입력하세요.

PERFORMANCE

Results

Initial Training

Trained with maximum images
3DNozzle / floor dust falsely
detected — frequent false
positives

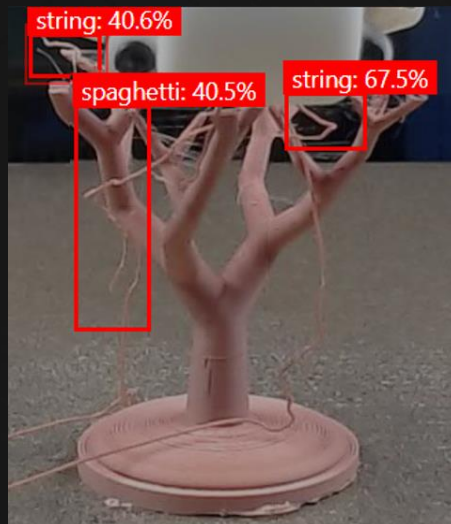
Revised Training Strategy

Reduced dataset size;
trained on clearly
distinctive images only

Outcome

Significant error reduction
(Quality over quantity matters)

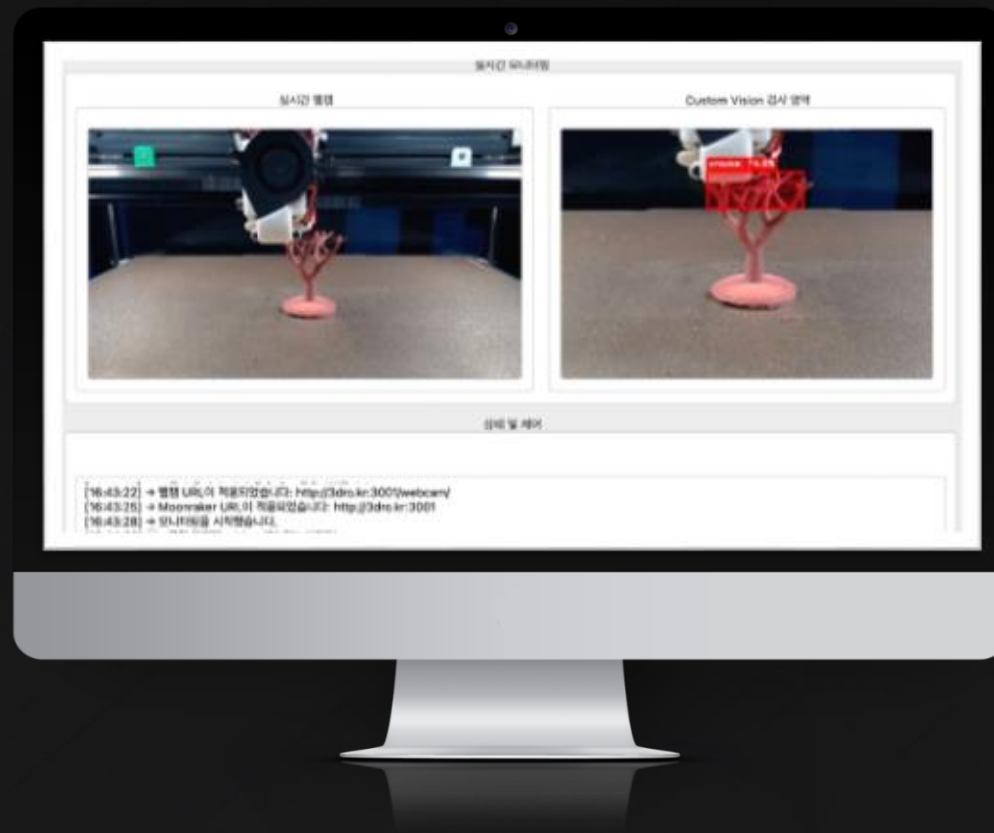
Final Test Result Images



PERFORMANCE

Results

Demo Video



모니터링 설정

Webcam URL: http://3dro.kr:3001/webcam/

적용

Moonraker URL: http://3dro.kr:3001

적용

웹캠 및 서버 링크 입력

실시간 웹캠



Custom Vision 검사 영역



상태 및 제어

[16:59:56] → 프린터 상태가 '대기 중'(으)로 변경되었습니다

▶ 모니터링 시작

■ 모니터링 중지

|| 일시정지

▶ 재개

AI ETHICS

AI Ethics Application

Privacy Protection & Security

- ✓ Terms of service **consent** provided
- ✓ **Azure Cloud security services for data storage**

Reliability

- ✓ **Accurately identifying print errors through continuous error detection algorithm improvement**

Accountability

- ✓ Errors are **logged** and traceable
- ✓ Customer support **system** activated for error handling

Fairness

- ✓ **Continuous data training and updates for diverse environment compatibility**

Inclusivity

- ✓ Program guides and support services provided
- ✓ Easy to use — **intuitive UI**

Transparency

- ✓ Program's **error identification criteria** and **notification standards** — **clearly communicated to users**

THANK YOU

Thank you!

Q & A